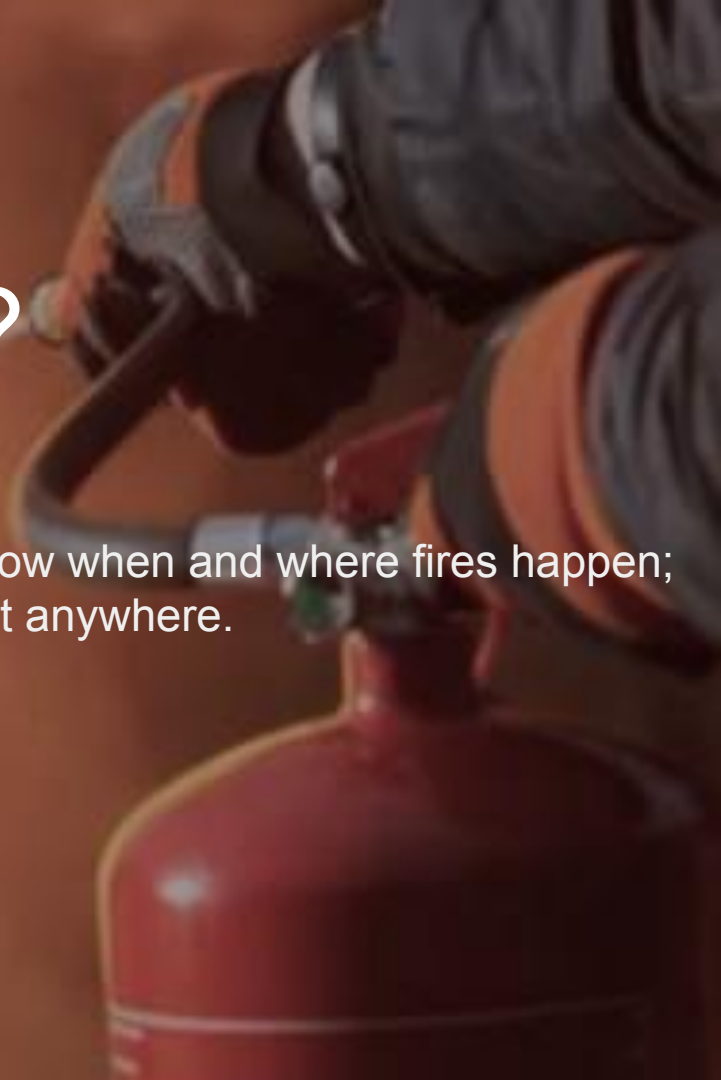


Hello! My name is Josiah Gipson

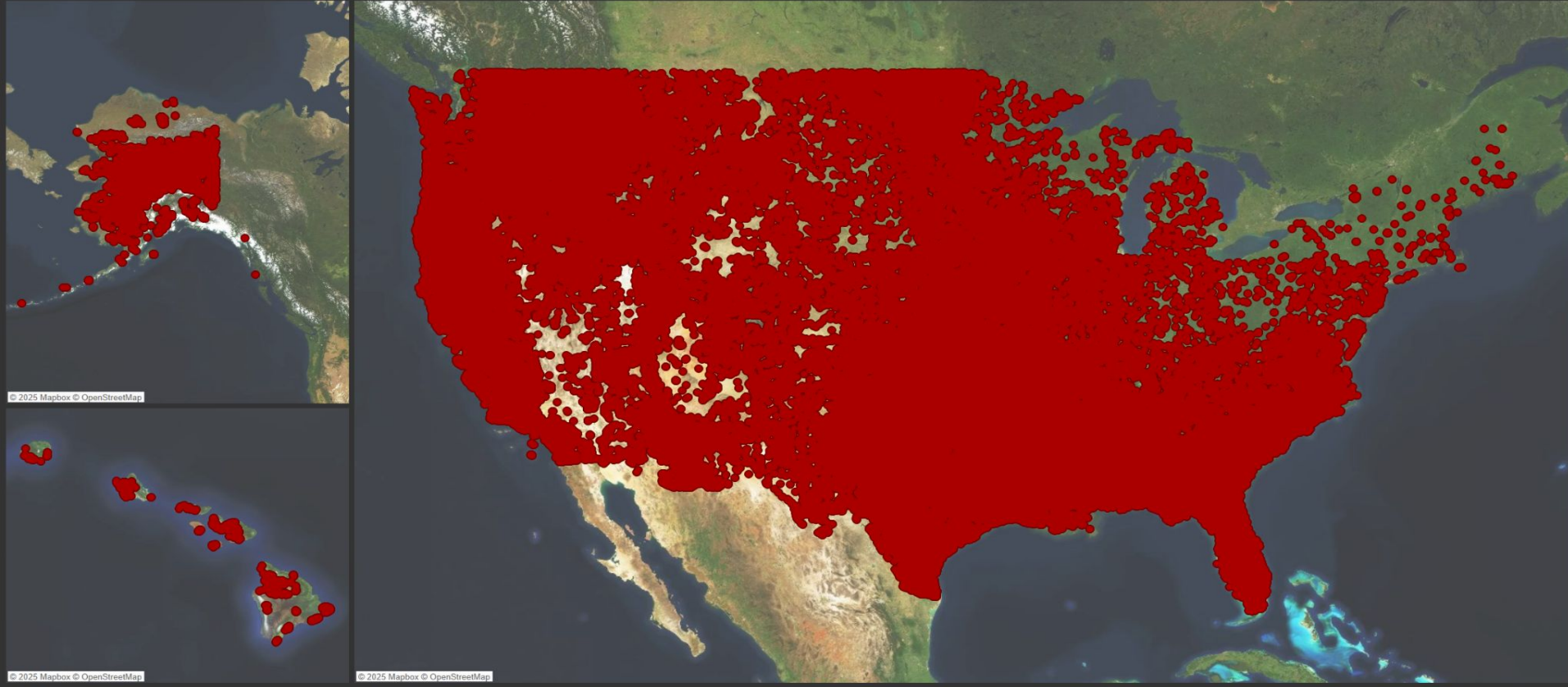
In this presentation I am going to
go over fire detection data in the US

Why fires?

Well fires can be dangerous, and it is always helpful to know when and where fires happen;
And I found they can happen almost anywhere.



Below you can see detected fires in the US since the year 2000
And it shows how almost nowhere is safe from fire.



A satellite image of Earth showing North America and surrounding regions. The image is dark, with the landmasses appearing in shades of green and brown against the deep blue of the oceans. White clouds are scattered across the scene, particularly concentrated over the western United States and the North Atlantic. The curvature of the Earth is visible on the left side of the frame.

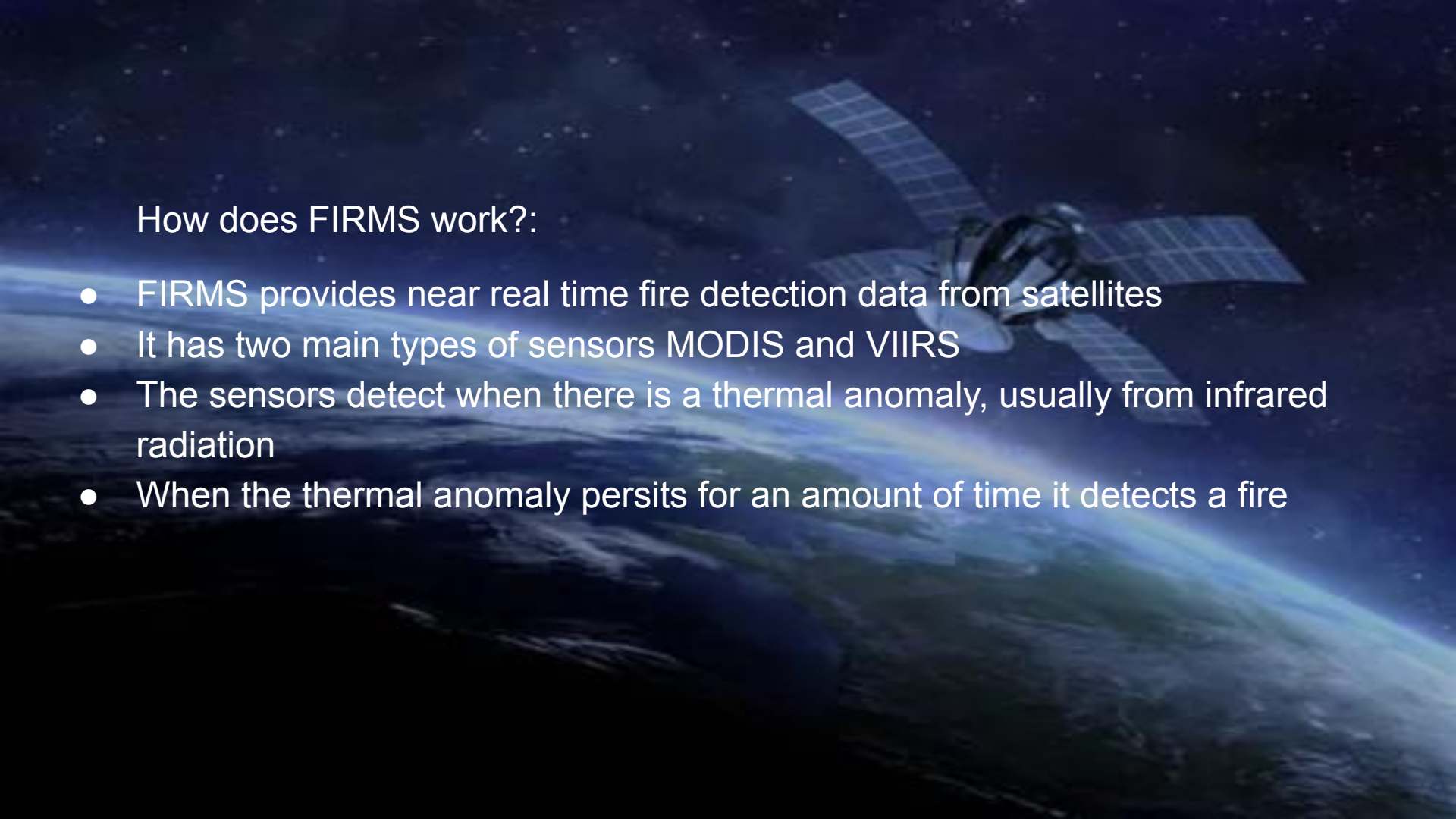
Now before I go into a deeper geographical and temporal analysis of where and when fires happen, I want to talk to you about FIRMS.



FIRMS stands for “Fire Information for Resource Management System”, and it is hosted by NASA. It is also the source for my data.

A satellite with four solar panels is shown in orbit above the Earth's cloud-covered surface. The satellite has a central body with a dish antenna and four rectangular solar panel arrays extending outwards. The Earth's horizon is visible with a thin blue line of atmosphere and white clouds. The background is the dark void of space filled with stars.

The data I used for this presentation was limited to the US, and has nearly 3 Million entries.

A satellite with large solar panels is shown in orbit above the Earth's atmosphere. The satellite is positioned in the upper right quadrant of the frame. The Earth's surface, showing clouds and land, is visible in the lower half of the image. The background is the dark void of space with some distant stars.

How does FIRMS work?:

- FIRMS provides near real time fire detection data from satellites
- It has two main types of sensors MODIS and VIIRS
- The sensors detect when there is a thermal anomaly, usually from infrared radiation
- When the thermal anomaly persists for an amount of time it detects a fire

A satellite with two large solar panel arrays is shown in orbit above the Earth's cloud-covered surface. The satellite's central body is metallic and complex, with various instruments and antennas visible. The background is the deep blue of space, filled with distant stars. The Earth's horizon is visible at the bottom, showing a thin layer of white clouds over a dark blue ocean or landmass.

When a fire is detected it is also associated with a confidence level that has a value from 0 to 100 which is grouped into High(greater than 80), Nominal/Normal(greater than 30), and Low(under 30), and I wanted to know what related to that confidence level, and if I could predict it based off of the other data in the dataset.



My methods

Models: Linear regression,
Logistic regression,
Polynomial features,
Random Forest Regressor,
Random Forest Classifier

Analysis: R2 score,
MSE(Mean square error),
Accuracy score,
Classification report,
Feature importances

What is the difference between models?

The Forests are essentially A collection of decision trees that compare the independent variables to set values. The Regressor is used for numerical values, while the Classifier is used for categorical values

The regression models use mathematical formulas to find the relation between independent and target variables, Linear is for numerical values, while Logistic is for categorical values.

What do the values mean?

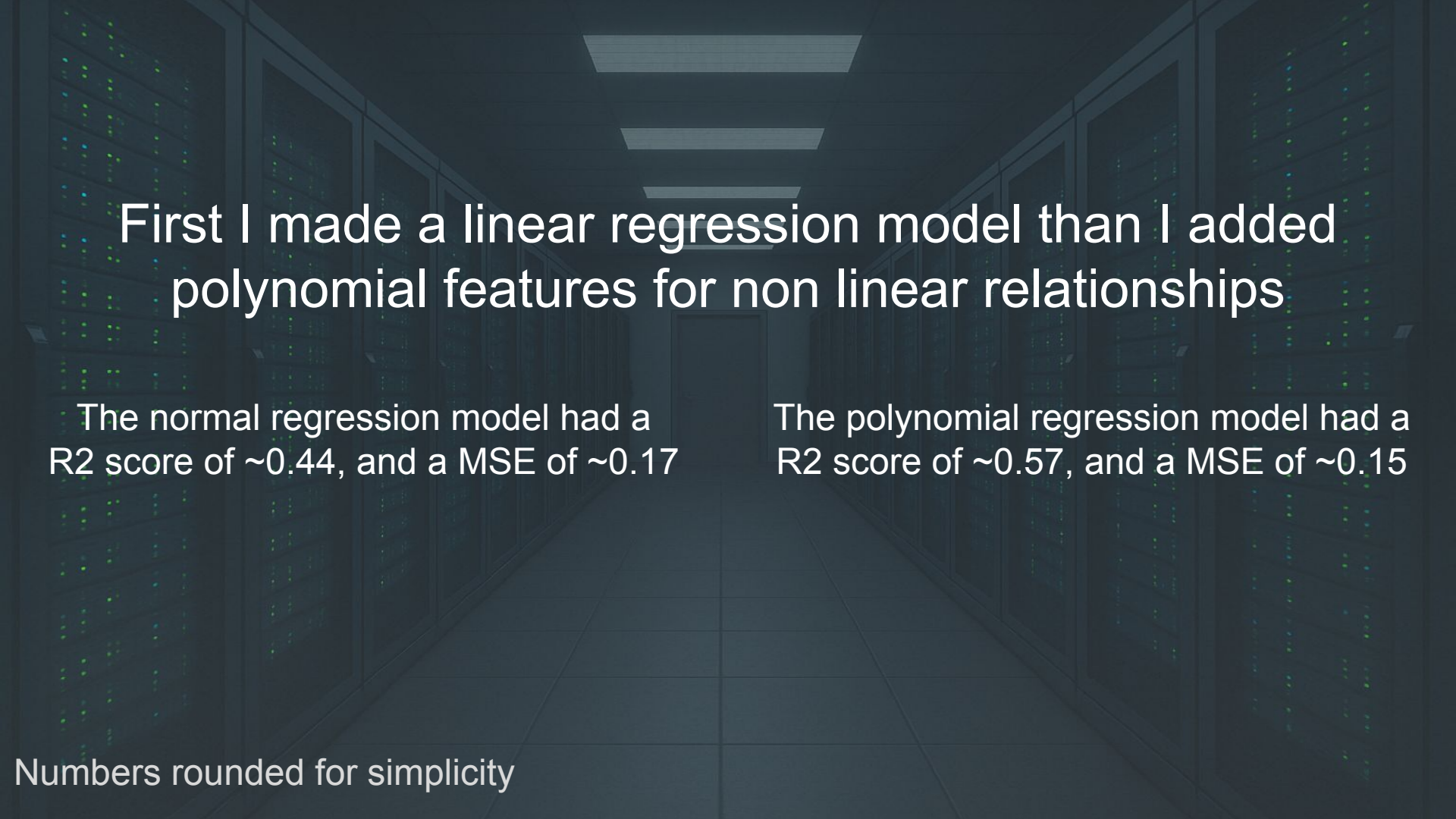
R2 score represents how well the model predicts,

MSE represents the average prediction error squared,

Precision is the ratio of correct positives over all predicted positives

Recall is the ratio of correct positives over all actual positives

f1-score is the average of precision and recall



First I made a linear regression model than I added polynomial features for non linear relationships

The normal regression model had a R2 score of ~ 0.44 , and a MSE of ~ 0.17

The polynomial regression model had a R2 score of ~ 0.57 , and a MSE of ~ 0.15

Numbers rounded for simplicity

Next I made a Random Forest Regressor model

The random forest regressor had a
R2 score of ~ 0.70 , and a MSE of ~ 0.13

Top feature importances

Brightness at 0.539, and bright_t31 at 0.060
latitude at 0.064, and longitude at 0.068
and frp(Fire Radiative Power) at 0.093

Numbers rounded for simplicity

Next I made a logistic regression and used polynomial features on logistic regression

Logistic regression

	precision	recall	f1-score	support
high	0.86	0.76	0.81	185330
nominal	0.81	0.94	0.87	355943
low	0.00	0.00	0.00	36574
accuracy			0.82	577847
Macro avg	0.56	0.57	0.56	577847
Weighted avg	0.79	0.82	0.80	577847

Polynomial Logistic regression

	precision	recall	f1-score	support
high	0.86	0.67	0.75	185524
nominal	0.78	0.95	0.86	355483
low	0.00	0.00	0.00	36840
accuracy			0.80	577847
Macro avg	0.55	0.54	0.54	577847
Weighted avg	0.76	0.80	0.77	577847

Numbers rounded for simplicity

And finally I made a Decision Tree Classifier

Decision tree

	precision	recall	f1-score	support
high	0.86	0.85	0.86	185618
nominal	0.87	0.86	0.86	355185
low	0.29	0.32	0.30	37044
accuracy			0.82	577847
Macro avg	0.67	0.68	0.67	577847
Weighted avg	0.83	0.82	0.83	577847

Top feature importances

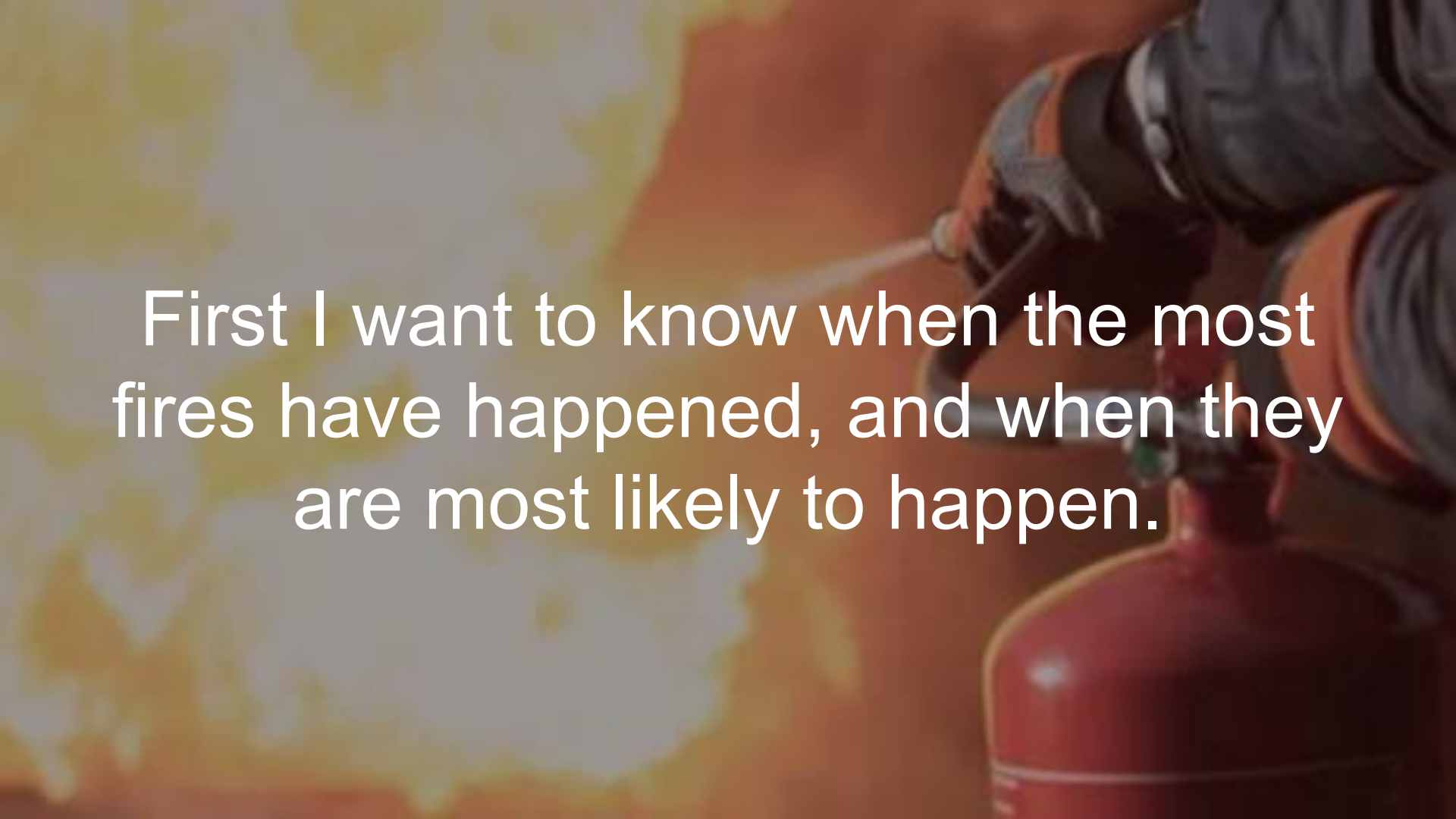
Brightness at 0.527, and bright_t31 at 0.074
latitude at 0.052, and longitude at 0.056
and frp(Fire Radiative Power) at 0.054

Numbers rounded for simplicity

Before we continue are there
any questions about the data or
the models I made?

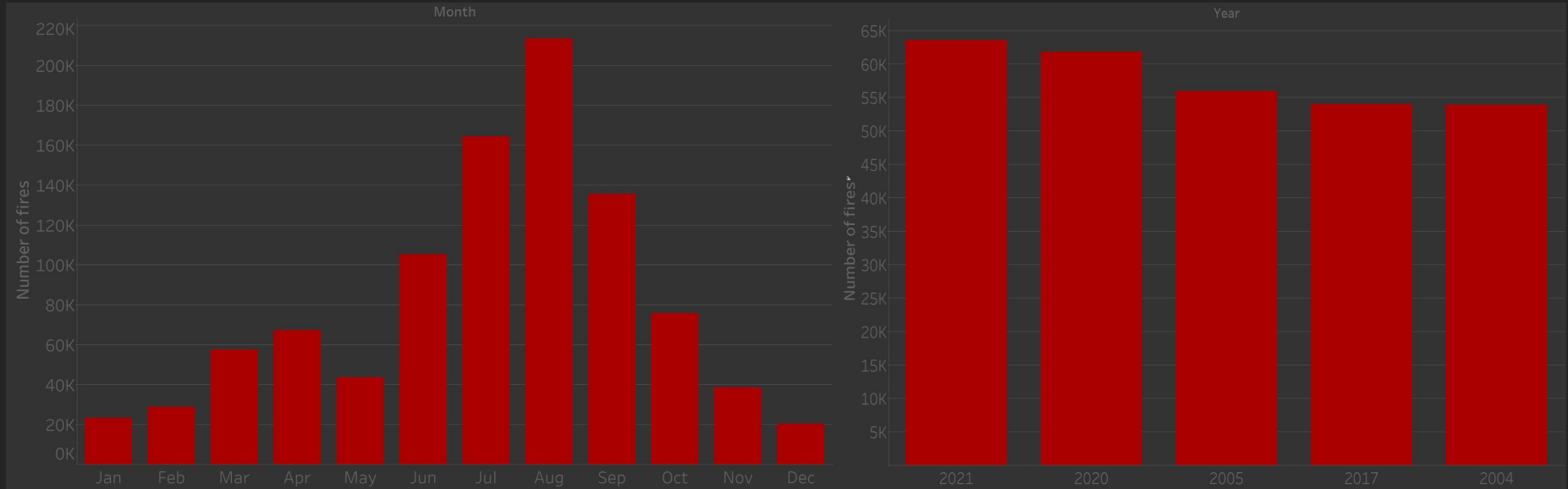
A satellite image of Earth showing North America, including the United States, Canada, and Mexico, surrounded by the Atlantic and Pacific Oceans. The image is used as a background for the text.

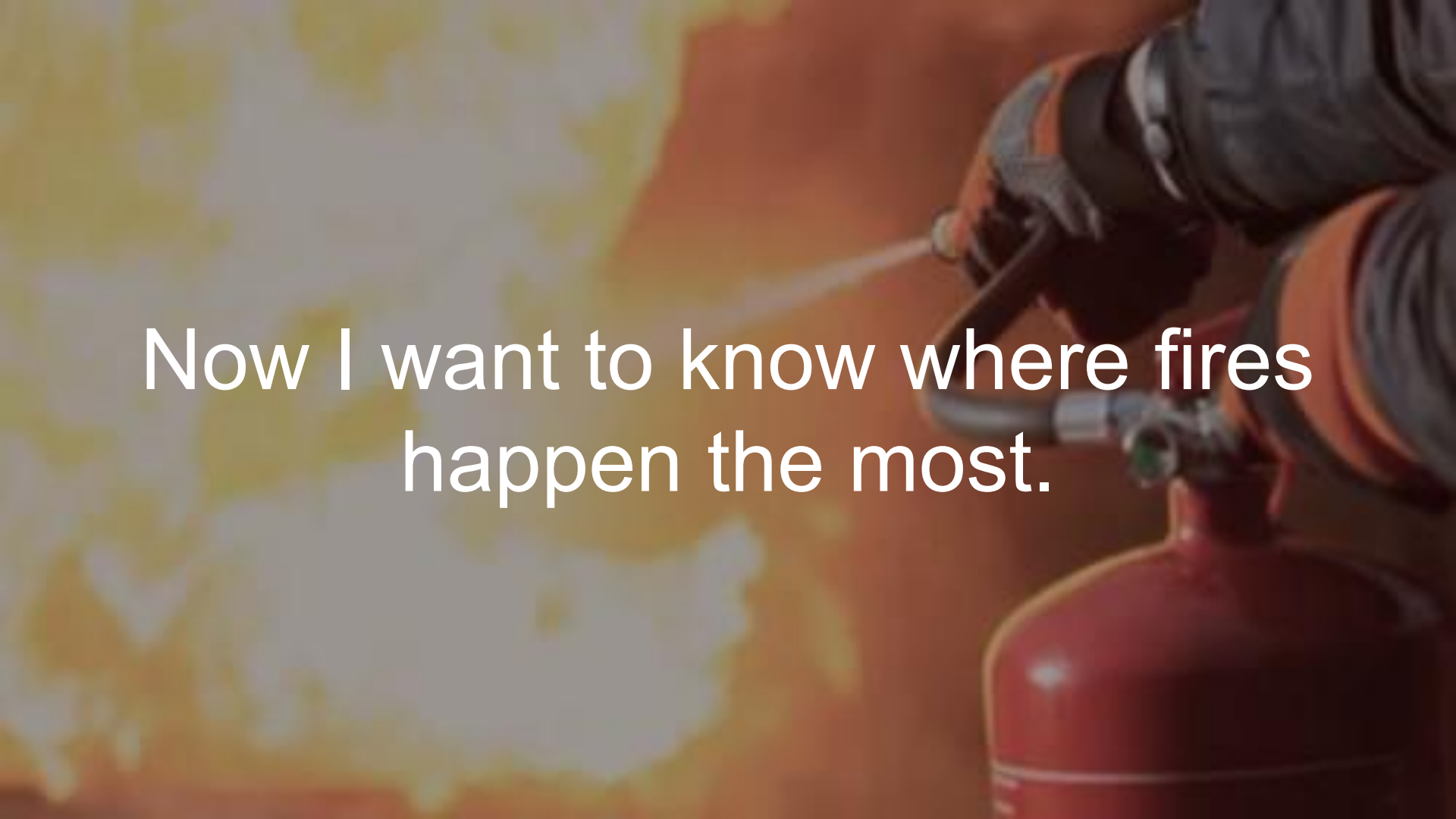
Now that the we know about the source of the data
let's go into geographical and temporal analysis



First I want to know when the most fires have happened, and when they are most likely to happen.

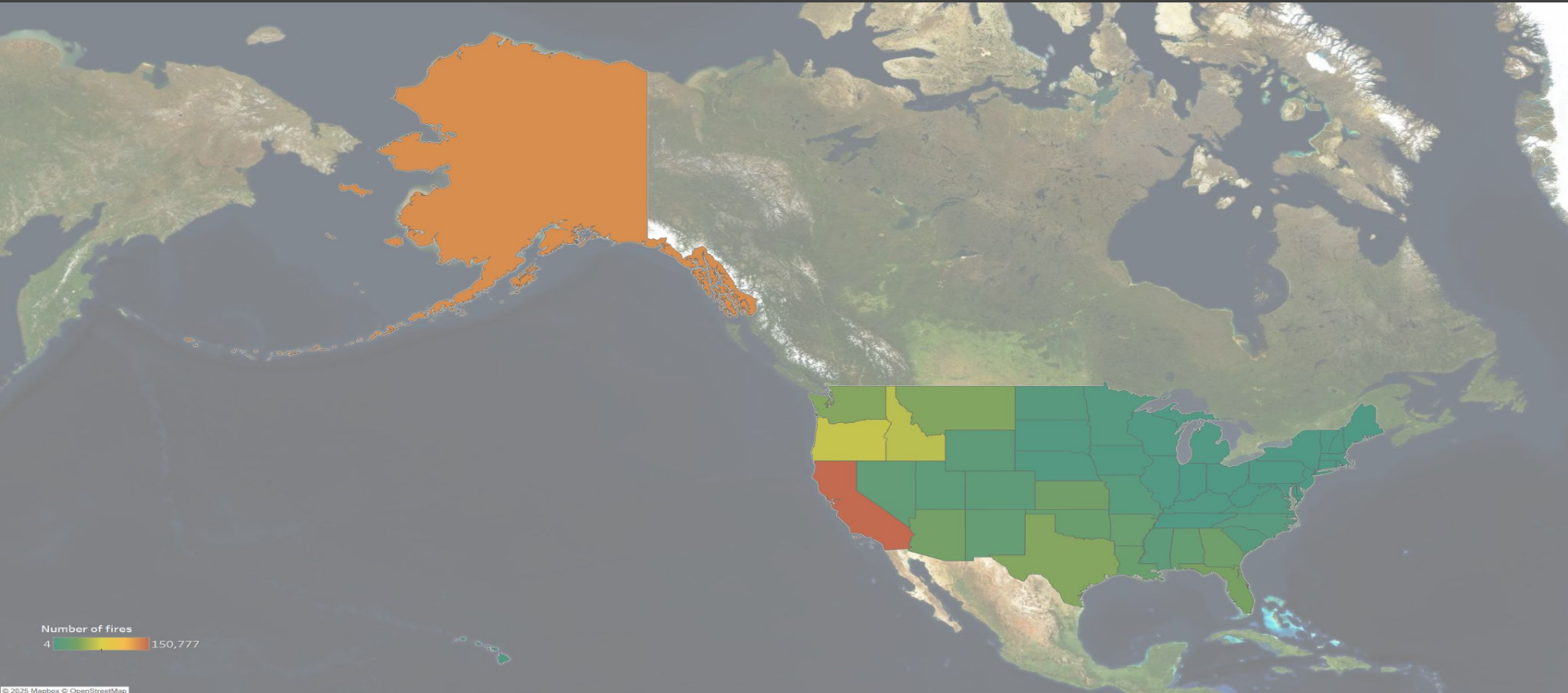
On the left you can see the number of fires for each month, and on the right you can see the top 5 years by number of fires.



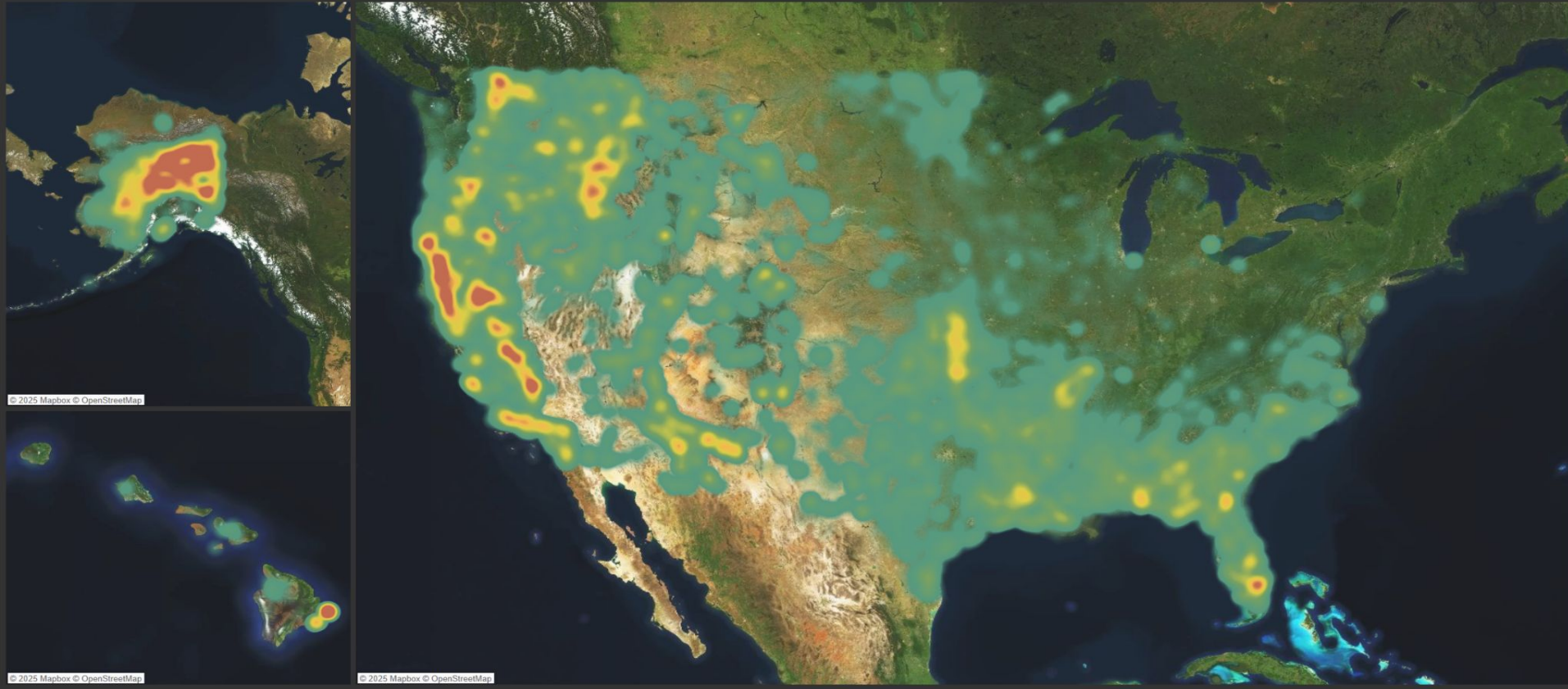


Now I want to know where fires
happen the most.

The graph below shows which states have had the most detected fires



Below is the heat map of fires since the year 2000



Summary

- FIRMS is a great source for knowing where and when fires happen
- The brightness of the fire is the most important value for confidence of the data from the dataset
- Fires happen the most in the summer and early fall months, and there isn't an outlier in the years recorded for number of fires
- Fires happen the most in Alaska and California